

WORD EMBEDDINGS

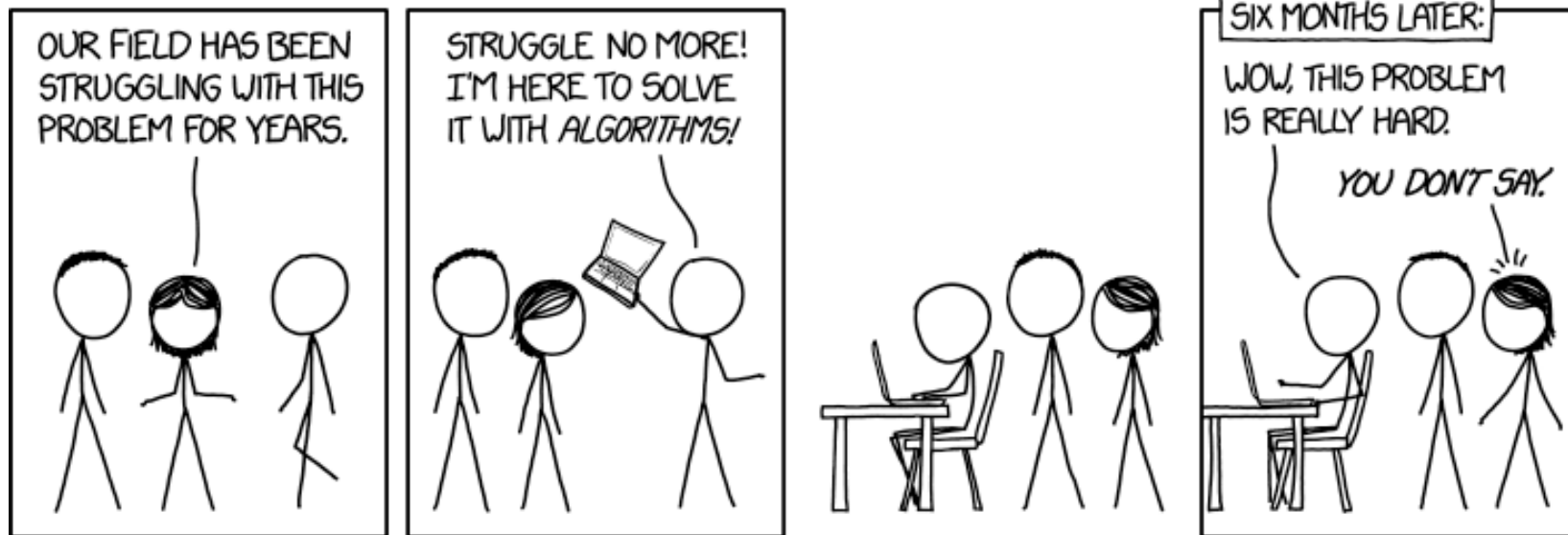
The heart of modern text analytics

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Working on difficult problems

Text analysts are getting good at solving specific problems. Our approach is to increase the specificity of the problem to improve the quality of result.



History of text analytics

3 phases of text analytics. Data / Computing availability has been a key driver for promoting the adoption of new approaches

**NLP / Programmed
Grammatical
Approaches**

Statistical Approaches

**Deep Learning / Word
Embeddings**

The key problem in text analytics:

Too many dimensions

‘Word-based NLP’ - Dimensionality Reduction techniques

Common techniques such as stop-word-removal, stemming and lemmatization all attempt to reduce dimensionality.

“My system keeps crashing! his crashed yesterday, ours crashes daily”

Stemming

**‘My system keep crash hi crash
yesterday, our crash daili’**

Lemmatization

**‘My system keep crash ! his crash
yesterday , ours crash daily’**

“You shall know a word by the company it keeps!”

John Firth

Page 11, “A synopsis of linguistic theory 1930-1955“, in *Studies in Linguistic Analysis 1930-1955*, 1962

Context is vital for understanding text information

Generic text algorithms fail to provide contextual meaning to comments. We look at text in 3 ways:

The context of the Organization

The context of the Question

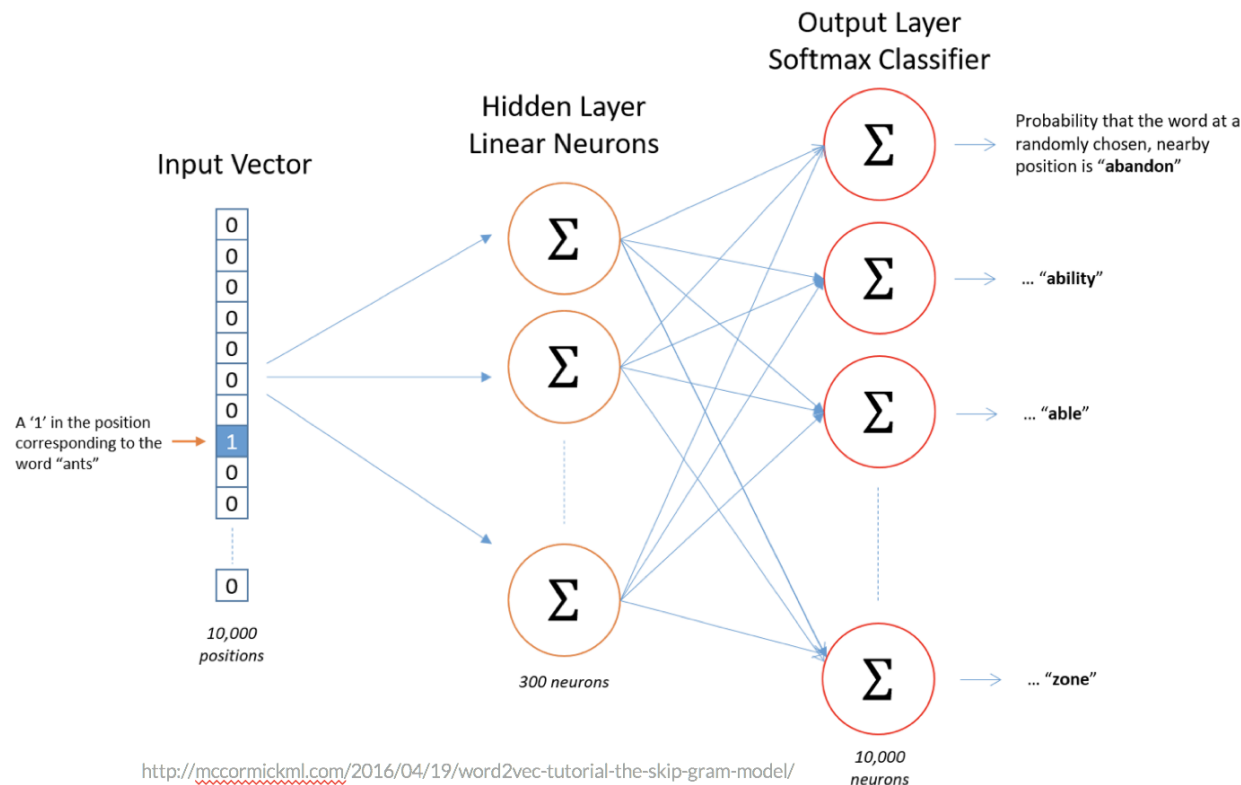
The context of the sentence structure

DEEP LEARNING & WORD EMBEDDINGS

Magic?

Word Embeddings - driving current growth

Word embedding approaches all use different approaches to understand words in their context.



“Skip-Gram”
With one-hot encoded centre word, we can predict context words.

Hidden layer creates embeddings

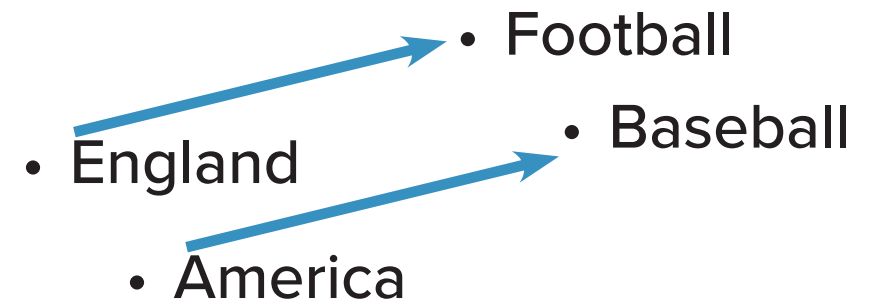
Word Embeddings

Dimensionality reduction can give you an idea of what is happening



Text embeddings learns relationships

Vectors can be subtracted to show learned relationships.



England	Football	America	Show	Top N: 2
Baseball America				

BMW	car	Boeing	Show	Top N: 2
airplane aircraft				

Word embeddings are everywhere

This looks a lot like a skip-gram model.



is st moritz **in switzerland**

where is st moritz

where is st moritz **on the map**

where is st moritz **hotel cornwall**

how is st moritz **pronounce**

where is st moritz **ski resort**

what canton is st moritz **in**

what altitude is st moritz

what lake is st moritz **on**

what is st moritz **bad**

Google Search

I'm Feeling Lucky

Report inappropriate predictions

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Text Embeddings

They can do some 'crazy' things

facebook research

Search



Research Areas

Publications

People

Academic Programs

Downloads & Projects

Careers

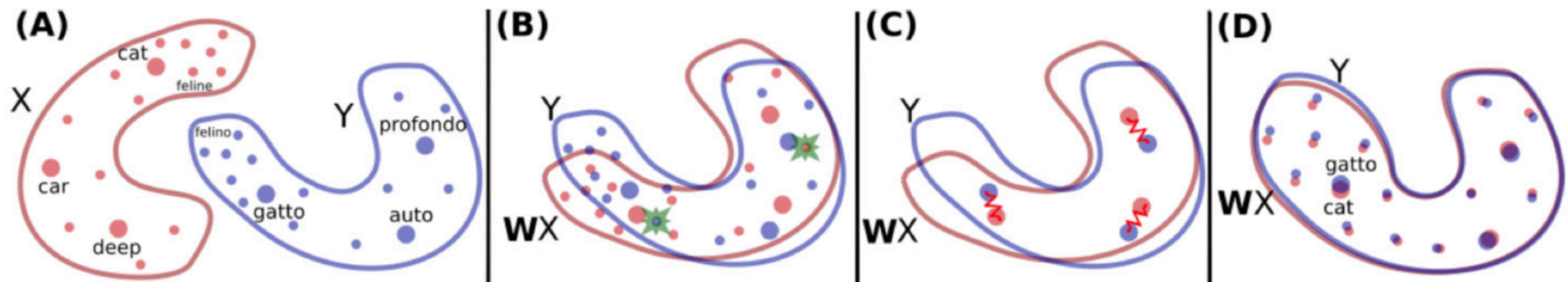
Blog

April 30, 2018

Word Translation Without Parallel Data

International Conference on Learning Representations (ICLR)

By: Alexis Conneau, Guillaume Lample, Marc'Aurelio Ranzato, Ludovic Denoyer, Hervé Jégou



Need to understand the semantic meaning

“If you could do one thing to improve working for COMPANY what would it be?”

“more money, less hours!”

**“Not being salary capped after
30 + years of loyal service”**

**“Competitive salary increases
for internal movement, low
pay rises encourage outward
moves rather than talent
retention”**

“Better pay”

“Better wages”

**“if I could I would give a pay
rise to the living wage amount
and not just the minimum
wage, we do work hard for our
company.”**

Need to understand the semantic meaning

“If you could do one thing to improve working for COMPANY what would it be?”

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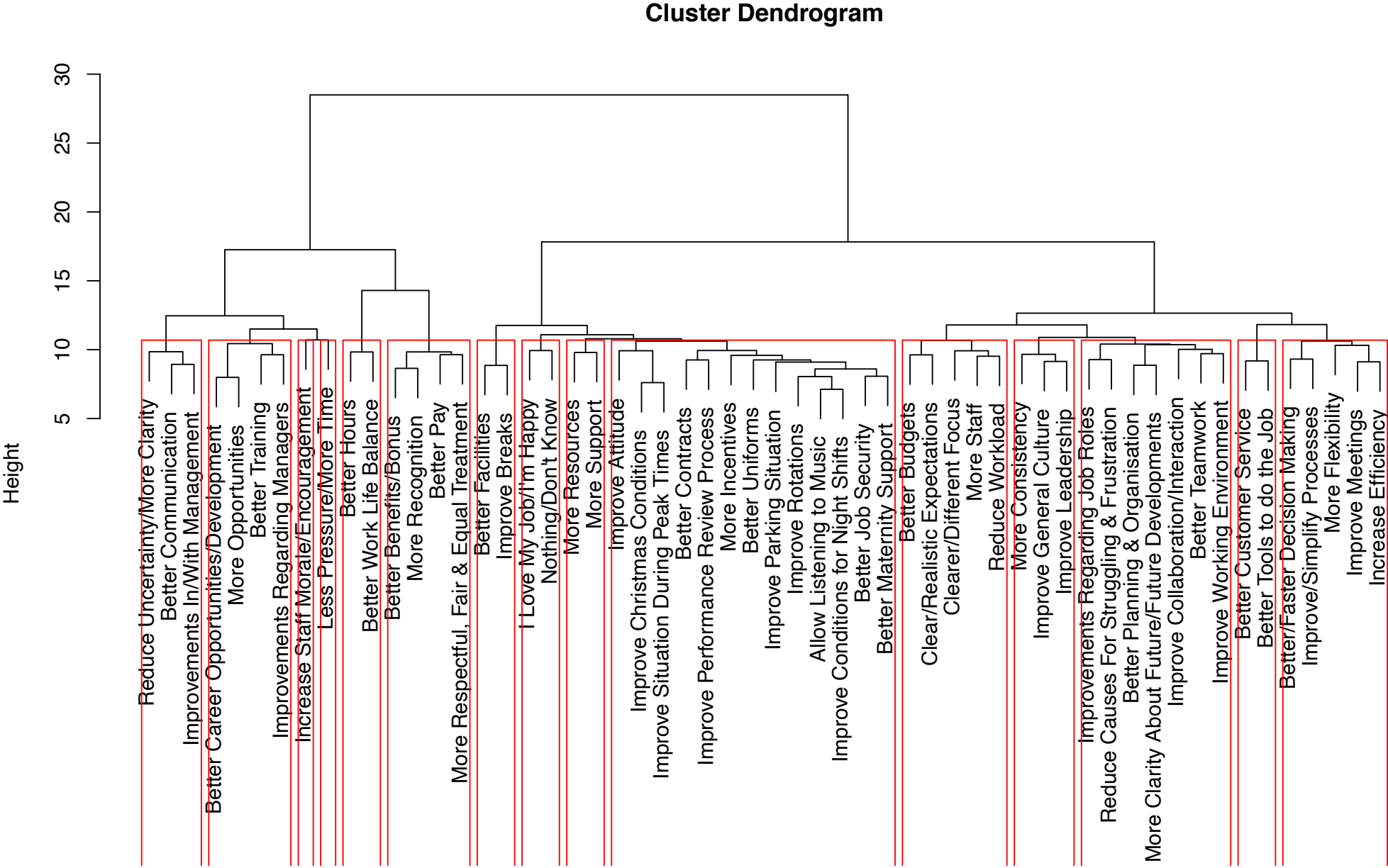
“Better pay”

“Ha Ha. A raise”

“Better wages”

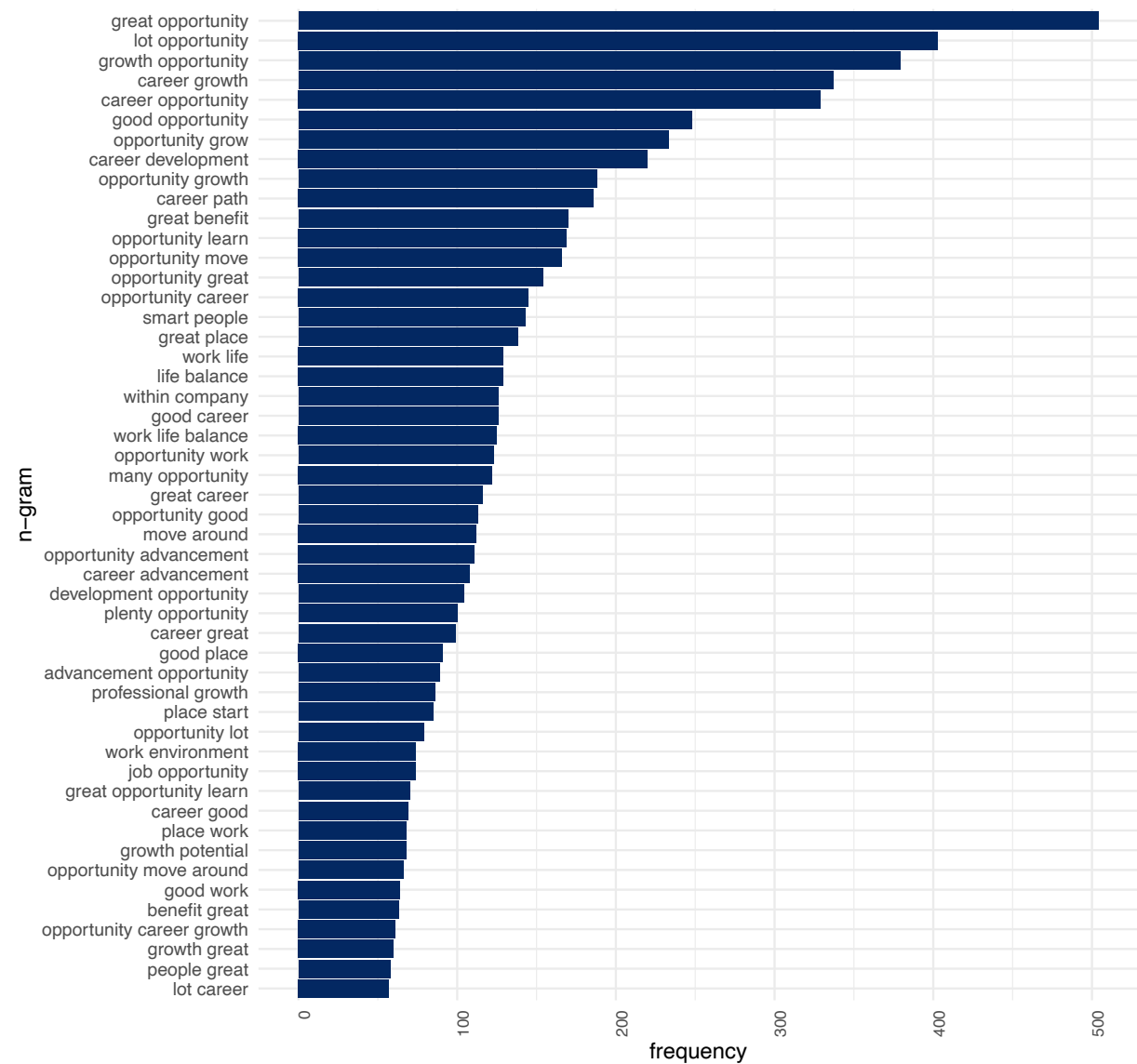
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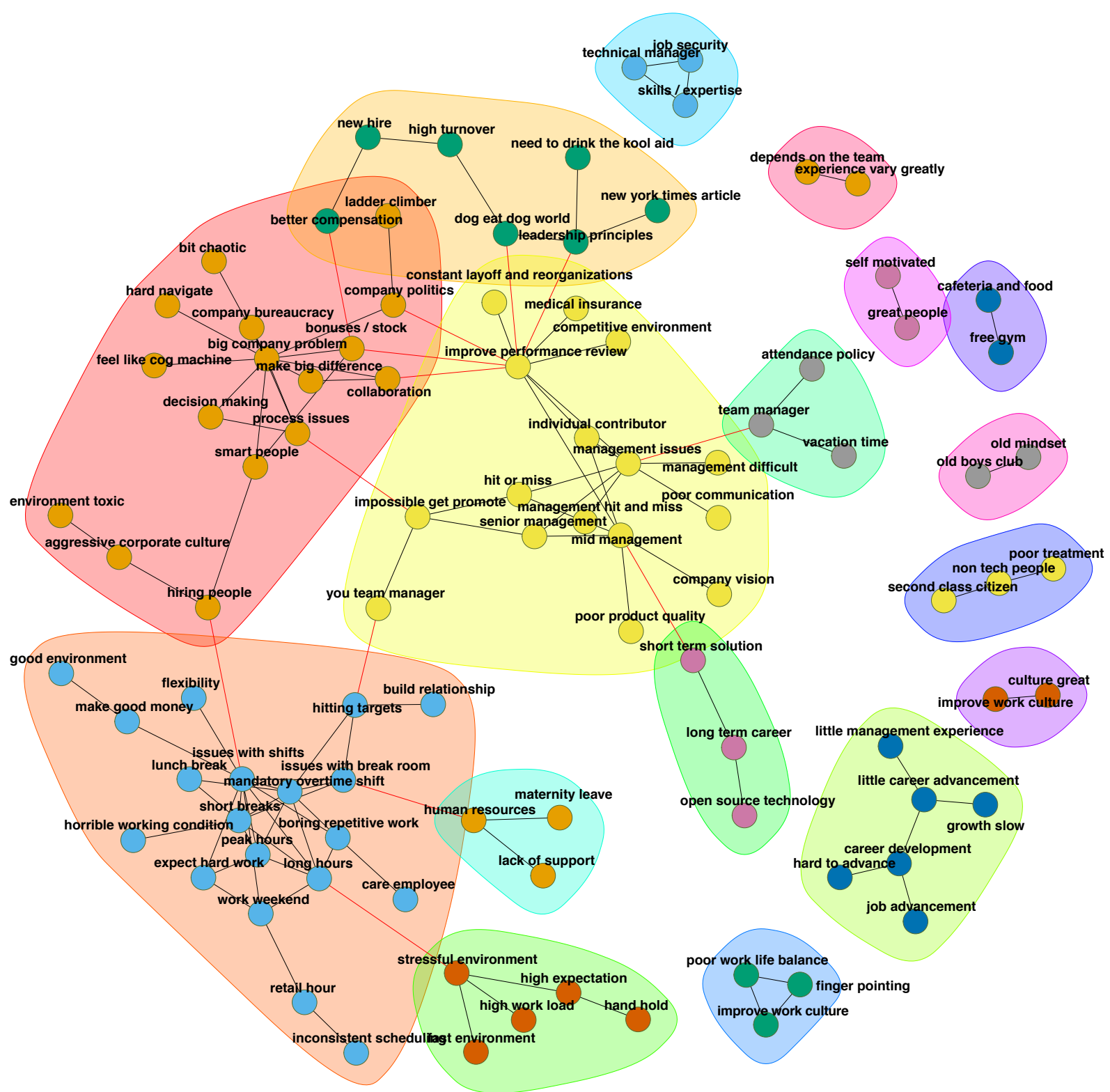
Clustering

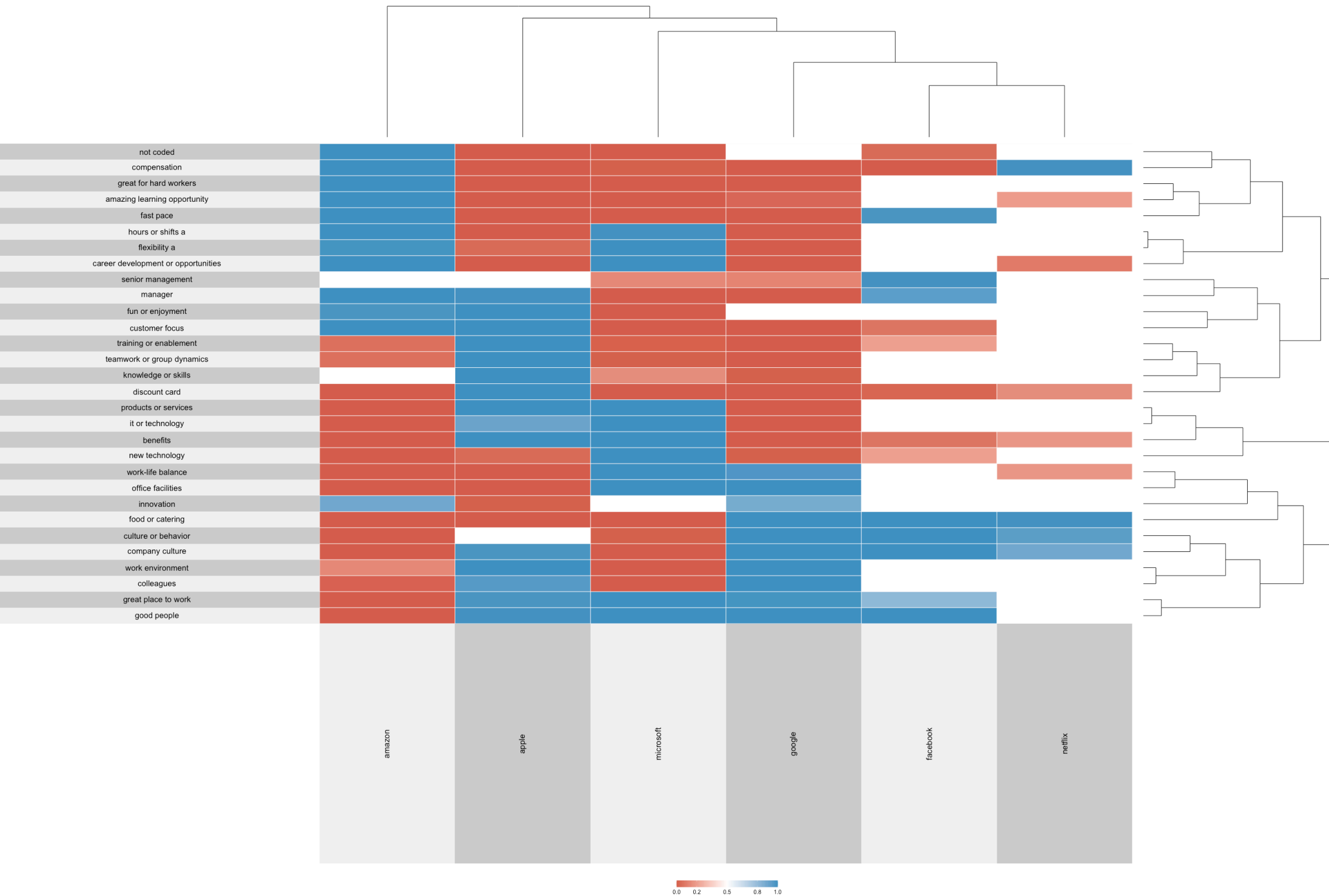


Reverse-engineering a topic

Dominant n-grams in career development or opportunities

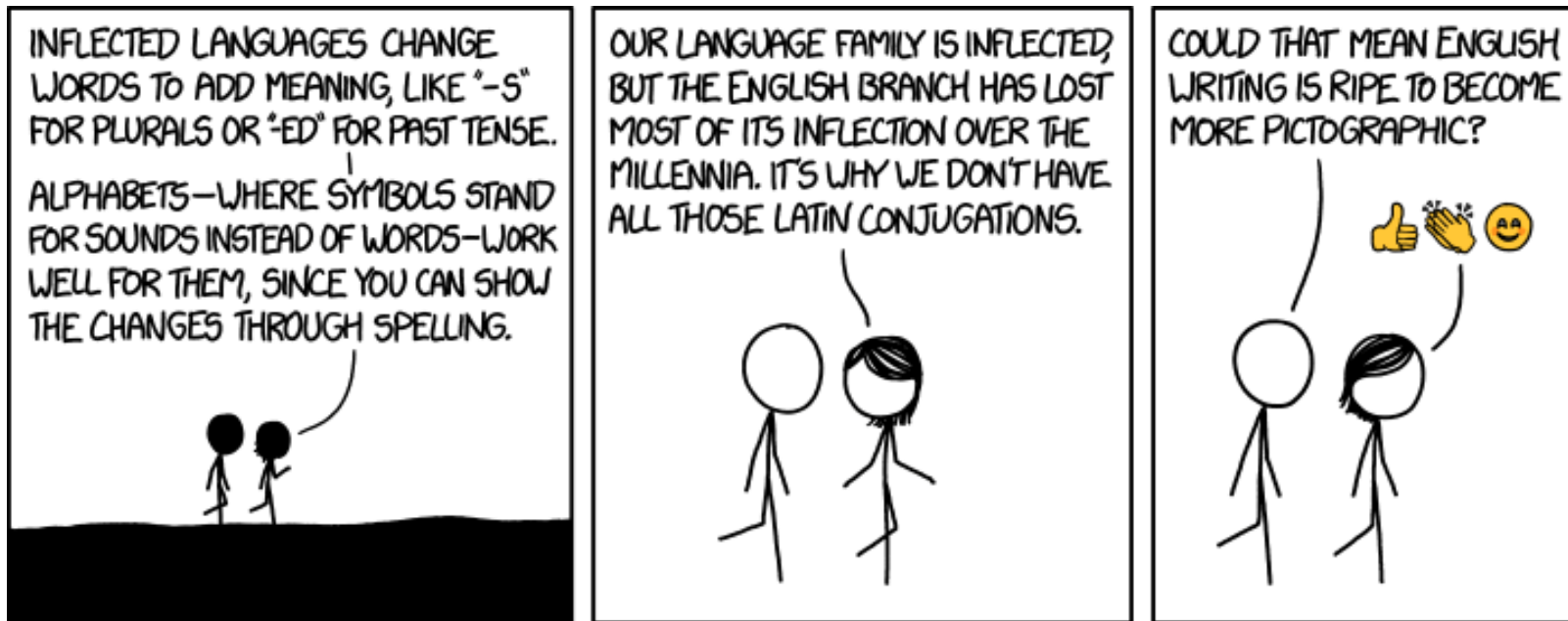






More than just understanding the words

We need to find a way to deal with the way people communicate, especially in short texts.



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